Module 1 — Synthesis Quiz

Question bank · After exercise lab · open notes · two attempts

Module 1, Synthesis quiz

Readings: Read 0, Read 1, Read 2
Labs: Demo, Exercise
Canvas: Synthesis quiz (Module 1) · **two attempts**, highest score counts

Full integration: binomial test workflow, **all Module 1 R practice**, plus interpretation items. Work sections in order — earlier sections match Read 2; R sections match the demo and exercise labs.

Instructions

- Open notes (reference sheets + methods map).
- **Graph items (Section C):** Run the R chunks to build each null plot; on your copy, add **`geom_vline` at observed** x and **shade the p-value region** matching H_a.
- Connect every step to Project 1.

Section A — Hypotheses and BRP (Read 2 §1)

A1. ESP hypotheses

ESP test: 25 trials, 5 symbols, guess each trial. Let p = probability of a correct guess on one trial.

1. Write H_0 and H_a in symbols.
2. In words: if H_0 is true? If we believe H_a?

A2. ESP BRP conditions

Same ESP setting (25 trials, open deck, one guess per trial). List the **four BRP conditions** and briefly say why each holds.

Answer key — Section A — Hypotheses and BRP (Read 2 §1)

1. $H_0 : p = 0.2$ vs $H_a : p > 0.2$. H_0: guessing only (no ESP). H_a: better than chance.
2. Binary (correct/incorrect); fixed $n = 25$; fixed $p = 0.2$ under null; independent trials (open deck).

Section B — n, p, and tail direction (Read 2 §2)

B1. n and p under H_0

ESP test (Part A): under H_0, what are n and p?

B2. Observed data and p-value direction

Same test; $H_0 : p = 0.2$ vs $H_a : p > 0.2$. You observe **7** correct out of 25.

1. What is x?
2. Write the p-value as $P(X \geq ...)$ or $P(X \leq ...)$.
3. Which tail(s) of the null distribution?

Answer key — Section B — n, p, and tail direction (Read 2 §2)

1. $n = 25$; $p = 0.2$.
2. $x = 7$; $P(X \geq 7)$; **right tail**.

Section C — Null plots: mark data and shade p-value (Read 2 §3)

```
if (!requireNamespace("ggplot2", quietly = TRUE)) {
  install.packages("ggplot2", repos = "https://cloud.r-project.org")
}
library(ggplot2)
```

Each plot below shows the **null distribution** (x vs $P(X = x)$ under H_0). Plots are **unshaded** — on your copy (print or sketch), add a **vertical line** at the observed x and **shade the p-value region** in the direction of H_a.

Run the R chunks to reproduce each null distribution in Colab or RStudio (same pattern as Project 1 and the demo lab).

C1. Left-handed students

Research question: Is the proportion of left-handed students at our school higher than the national rate of 10%?

Parameter: p = proportion of students at our school who are left-handed.

Hypotheses: $H_0 : p = 0.10$ vs $H_a : p > 0.10$. Under H_0, use $n = 25$ and the null value of p shown in the R chunk.

Data: observed $x = 6$.

```
n <- 25
p <- 0.10
null_df <- data.frame(x = 0:n, prob = dbinom(0:n, size = n, prob = p))
ggplot(null_df, aes(x = x, y = prob)) +
  geom_col() +
  labs(x = "x", y = expression(P(X == x))) +
  theme_minimal()
```

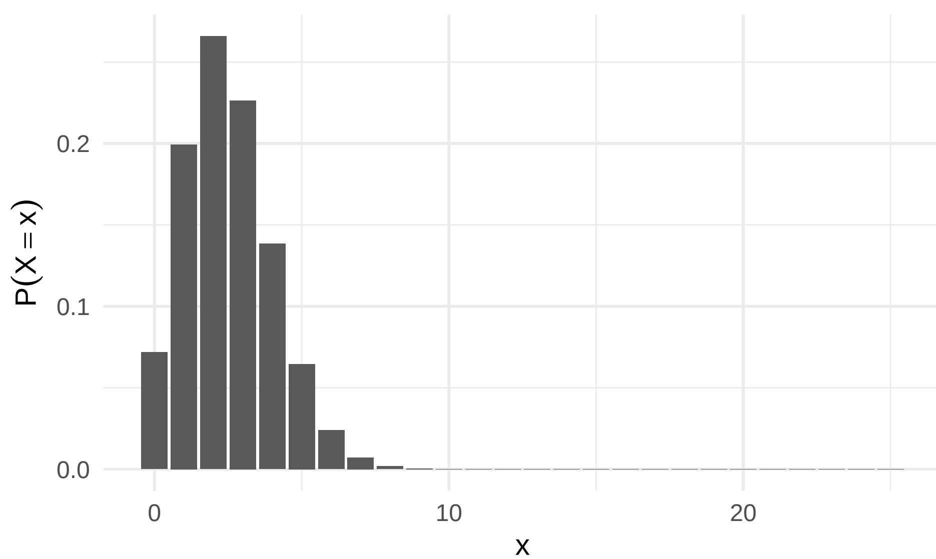

Figure 1: Null distribution: left-handed students, n = 25, p = 0.10 under H0

1. On the plot (or a printout), add a **vertical line** at $x = 6$ and **shade** the p-value region (right tail).
2. p-value $= P(X \geq 6 \mid H_0)$ is **small** or **large**?
3. Conclusion in context (one sentence).

C2. Defective items

Research question: Is the proportion of defective items from the new supplier less than 15%?

Parameter: p = proportion of items from the new supplier that are defective.

Hypotheses: $H_0 : p = 0.15$ vs $H_a : p < 0.15$. Under H_0, use $n = 25$ and the null value of p shown in the R chunk.

Data: observed $x = 3$.

```r
n <- 25
p <- 0.15
null_df <- data.frame(x = 0:n, prob = dbinom(0:n, size = n, prob = p))
ggplot(null_df, aes(x = x, y = prob)) +
  geom_col() +
  labs(x = "x", y = expression(P(X == x))) +
  theme_minimal()
```

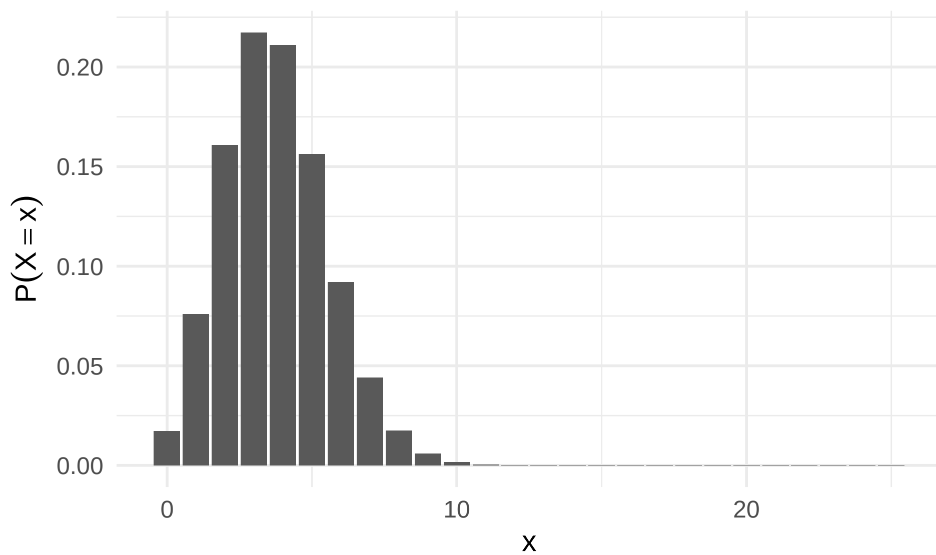

Figure 2: Null distribution: defective items, n = 25, p = 0.15 under H0

1. On the plot (or a printout), add a **vertical line** at $x = 3$ and **shade** the p-value region (left tail).
2. p-value $= P(X \leq 3 \mid H_0)$ is **small** or **large**?
3. Conclusion in context (one sentence).

C3. Dice (sum 7 or 11)

Research question: When we roll two dice, is the probability that the sum is 7 or 11 equal to the fair-dice value (8/36)?

Parameter: p = probability that one roll of two dice gives sum 7 or 11.

Hypotheses: $H_0 : p = 8/36$ vs $H_a : p \neq 8/36$. Under H_0, use $n = 25$ and the null value of p shown in the R chunk.

Data: observed $x = 2$.

```r
n <- 25
p <- 8/36
```

```r
null_df <- data.frame(x = 0:n, prob = dbinom(0:n, size = n, prob = p))
ggplot(null_df, aes(x = x, y = prob)) +
  geom_col() +
  labs(x = "x", y = expression(P(X == x))) +
  theme_minimal()
```

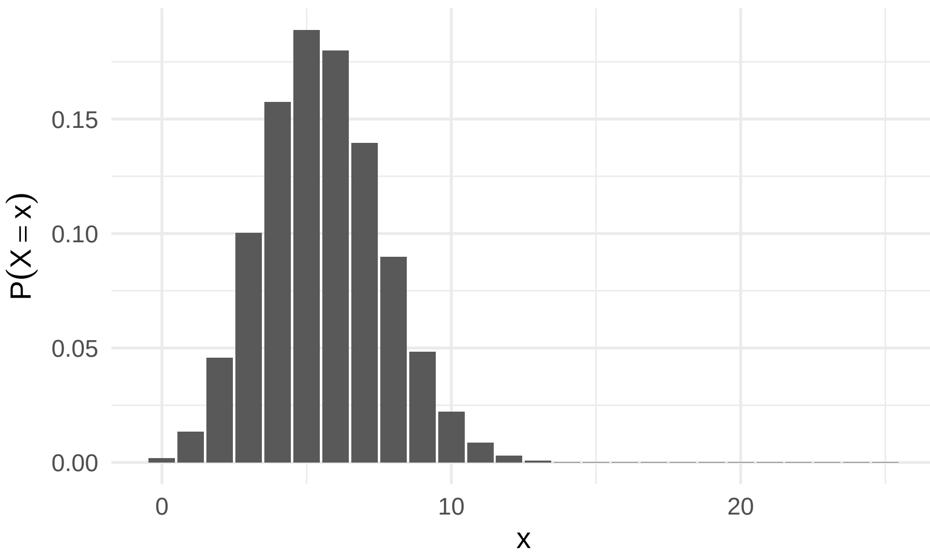

Figure 3: Null distribution: dice sum 7 or 11, n = 25, p = 8/36 under H0

1. On the plot (or a printout), add a **vertical line** at $x = 2$ and **shade** the p-value region (both tails (two-sided)).
2. p-value $= P(X \leq 2) + P(X \geq 9)$ under H_0 is **small** or **large**?
3. Conclusion in context (one sentence).

C4. Gym use

Research question: Do more than 20% of students at our school use the gym at least once per week?

Parameter: p = proportion of students who use the gym at least once per week.

Hypotheses: $H_0 : p = 0.20$ vs $H_a : p > 0.20$. Under H_0, use $n = 25$ and the null value of p shown in the R chunk.

Data: observed $x = 5$.

```r
n <- 25
p <- 0.20
null_df <- data.frame(x = 0:n, prob = dbinom(0:n, size = n, prob = p))
ggplot(null_df, aes(x = x, y = prob)) +
  geom_col() +
  labs(x = "x", y = expression(P(X == x))) +
  theme_minimal()
```

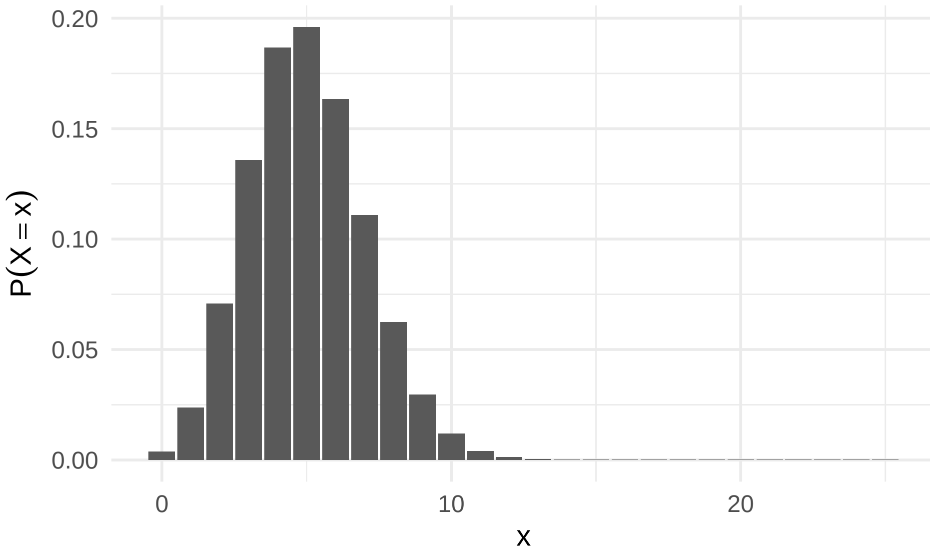

Figure 4: Null distribution: gym use, n = 25, p = 0.20 under H0

1. On the plot (or a printout), add a **vertical line** at $x = 5$ and **shade** the p-value region (right tail).
2. p-value $= P(X \geq 5 \mid H_0)$ is **small** or **large**?
3. Conclusion in context (one sentence).

Answer key — Section C — Null plots: mark data and shade p-value (Read 2 §3)

1. Shade bars $x \geq 6$; **small** (about 0.04); reject H_0 — evidence proportion at school **exceeds** 10%.
2. Shade $x \leq 3$; **large** (about 0.5); **fail to reject** — data consistent with 15% defect rate.
3. Shade $x \leq 2$ and $x \geq 9$; typically **large**; **fail to reject** fair-dice proportion.
4. Shade $x \geq 5$; **large** (about 0.5); **fail to reject** — no strong evidence share exceeds 20%.

Section D — Test workflow and one- vs two-sided (Read 2 §3)

D1. Order the steps

Number these binomial-test steps **1–4** in order:

- **(A)** Verify the four BRP conditions.
- **(B)** State research question, parameter p, H_0, H_a, and n, p under H_0.
- **(C)** Plot null distribution (x vs $P(X = x)$ under H_0).
- **(D)** Collect x, add observed value to plot, shade p-value region, compute p-value, conclude.

D2. One- vs two-sided shading

1. One-sided H_a (e.g. $p > 0.2$): shade **one tail** or **both tails**?
2. Two-sided H_a (e.g. $p \neq 8/36$): shade **one tail** or **both tails**?

Answer key — Section D — Test workflow and one- vs two-sided (Read 2 §3)

1. 1=B, 2=A, 3=C, 4=D.
2. 1. **One tail** (direction of H_a). 2. **Both tails**.

Section E — Binomial R and ggplot (after demo lab)

E1. `dbinom`

One line: $P(X = 5)$ when $X \sim \text{Binomial}(10, 0.4)$. Store in `prob_five`.

E2. `pbinom` lower tail

One line: $P(X \leq 3)$ for the same distribution. Store in `prob_at_most_3`.

E3. `pbinom` upper tail

One line: $P(X > 2)$ using `lower.tail = FALSE`. Store in `prob_more_than_2`.

E4. `rbinom`

One line: simulate 1000 values from $\text{Binomial}(8, 0.2)$. Store in `sims`.

E5. Estimate from simulation

Using `sims` from E4, one line to estimate $P(X \leq 2)$. Store in `p_est`.

E6. `aes()` in ggplot2

What does `aes()` do? Choose the best answer:

a) Creates the axes.

b) Maps variables to visual properties (x, y, color, fill).

c) Chooses the geometry (`geom_col`, etc.).

d) Defines the data frame.

E7. Null bar plot with `geom_col`

Fill in blanks using: `0:10` | `dbinom(0:10, size = 10, prob = 0.4)` | `x` | `prob`

```
df <- data.frame(x = _______, prob = _______)
ggplot(df, aes(x = _______, y = _______)) + geom_col()
```

Answer key — Section E — Binomial R and ggplot (after demo lab)

1. `prob_five <- dbinom(5, size = 10, prob = 0.4)`
2. `prob_at_most_3 <- pbinom(3, size = 10, prob = 0.4)`
3. `prob_more_than_2 <- pbinom(2, size = 10, prob = 0.4, lower.tail = FALSE)`
4. `sims <- rbinom(1000, size = 8, prob = 0.2)`
5. `p_est <- mean(sims <= 2)`
6. **(b)** — maps variables to visual properties.
7. `df <- data.frame(x = 0:10, prob = dbinom(0:10, size = 10, prob = 0.4))` then `aes(x = x, y = prob)`.

Section F — R basics (Module 0 demo / exercise)

F1. Arithmetic

One line of R that produces **10** by adding two numbers.

F2. Multiplication

One line of R that produces **6** by multiplying two numbers.

F3. Assignment and `sqrt`

Assign **16** to `x`, then compute $\sqrt{x}$.

F4. Vector

Create vector `numbers` with values 2, 4, 6, 8, 10.

F5. Indexing

Using `numbers`, return the **third** element.

F6. Vector arithmetic

Create `numbers_plus_one` by adding 1 to every element of `numbers`.

F7. Mean

Compute the mean of `numbers`.

F8. Count with `==` and `sum`

Given `flips <- c("H","T","H","H","T","H")` and `heads <- c("H","H","H","H","H","H")`, count how many `"H"` appear using `==` and `sum()`.

Answer key — Section F — R basics (Module 0 demo / exercise)

1. Any valid addition, e.g. `3 + 7` or `5 + 5`.
2. e.g. `2 * 3`.
3. `x <- 16` then `sqrt(x)` (two lines or one).
4. `numbers <- c(2, 4, 6, 8, 10)`
5. `numbers[3]`
6. `numbers_plus_one <- numbers + 1`
7. `mean(numbers)`
8. `sum(flips == heads)` or `sum(flips == "H")` → 4.

Section G — Vectors, matrices, loops (demo Section B / exercise Part I)

G1. Types and `class()`

Create `x_num <- 0`, `x_int <- 0L`, and print the class of each.

G2. Vector and length

Create `v <- c(2,4,6,8,10)`, print `length(v)`, create `v_plus_one <- v + 1`.

G3. Bracketing

With `v <- c(2,4,6,8,10)`, write code for: (1) 3rd element; (2) elements 2–4; (3) all except 3rd; (4) elements > 5.

G4. Matrix and `dim()`

Create 2×3 matrix `m` from `1:6` (by column), print `dim(m)`, extract row 2 col 1, first row, second column.

G5. `for` loop squares

Use a `for` loop so `squares` contains $1^2, 2^2, \ldots, 5^2$.

G6. Simulate P(even die)

Estimate P(even die) with `n <- 10000`, `rolls <- sample(1:6, size=n, replace=TRUE)`, and `mean(...)`. Name objects `n`, `rolls`, `p_even`.

Answer key — Section G — Vectors, matrices, loops (demo Section B / exercise Part I)

1. `x_num <- 0`; `x_int <- 0L`; `class(x_num)`; `class(x_int)`.
2. As stated; `length(v)` returns 5.
3. `v[3]`; `v[2:4]`; `v[-3]`; `v[v > 5]`.
4. `m <- matrix(1:6, nrow=2, ncol=3)`; `dim(m)`; `m[2,1]`; `m[1,]`; `m[,2]`.
5. `squares <- numeric(5)`; `for (i in 1:5) squares[i] <- i^2` (or `squares <- i^2` inside loop).
6. `p_even <- mean(rolls %% 2 == 0)` or `mean(rolls %in% c(2,4,6))`.

Section H — Simulation R (conditional demo / exercise Part II)

H1. `rep()` for balls

Create vector `balls`: 5 red and 2 white, labels `"R"` and `"W"`.

H2. Simulate without replacement

Bag: 3 red, 3 white. Simulate 10,000 draws of 2 without replacement; success = white first, red second; estimate with `mean(successes)`.

H3. `which()`

`successes` is length 10,000 (1=success, 0=failure). One line: which replicate numbers had success?

H4. Modify `if` for exactly one red

Change the condition so the loop estimates P(exactly one red) — (R,W) or (W,R) — using `|`.

H5. `sample()` with `prob`

Simulate 10,000 people: 70% like eggnog, 30% do not. Use `prob` in `sample()`; store in `eggnog`.

H6. Joint probability from simulation

Vectors `disease` and `test_result` (0/1). One line to estimate P(disease and positive) using `sum()` and `&`.

Answer key — Section H — Simulation R (conditional demo / exercise Part II)

1. `balls <- rep(c("R","W"), c(5,2))`
2. Loop or replicate with `sample(balls, 2, replace=FALSE)` and indicator; `mean(successes)` 1/7.
3. `which(successes == 1)`
4. Replace `if` with e.g. `if (sum(draw == "R") == 1)` or `(draw[1]=="R" & draw[2]=="W") | (draw[1]=="W" & draw[2]=="R")`.
5. `eggnog <- sample(c("yes","no"), 10000, replace=TRUE, prob=c(0.7,0.3))`
6. `sum(disease == 1 & test_result == 1) / length(disease)` or `mean(disease == 1 & test_result == 1)`.

Section I — Assumptions and interpretation (new)

I1. p-value meaning

Which statement is **incorrect**?

a) The p-value assumes H_0 is true when we compute it.

b) A small p-value means the observed data would be rare if H_0 were true.

c) The p-value is the probability that H_0 is true.

d) We shade tail(s) matching the direction of H_a.

I2. Which BRP condition fails?

You flip the same coin 25 times but the coin is **bent more after each flip** (success probability changes). Which BRP condition fails?

I3. Choosing `pbinom` for $H_a : p > 0.2$

Observed $x = 8$, $n = 25$, $p = 0.2$ under H_0, $H_a : p > 0.2$. Which line gives the p-value?

a) `pbinom(8, 25, 0.2)`

b) `pbinom(7, 25, 0.2, lower.tail=FALSE)`

c) `pbinom(8, 25, 0.2, lower.tail=FALSE)`

I4. Purpose of `geom_vline`

On a null binomial bar plot, what does `geom_vline(xintercept = x_obs)` show?

I5. `dbinom` vs `pbinom`

When plotting the null distribution with `geom_col`, which function gives bar **heights** $P(X = x)$ for each x?

I6. Conclusion language

p-value is 0.03 for $H_a : p > 0.10$. Write one sentence conclusion **in context** (not "accept/reject" jargon alone).

Answer key — Section I — Assumptions and interpretation (new)

1. **(c)** is incorrect — p-value is about **data** extremeness, not $P(H_0$ true).
2. **Fixed** p — success probability is not the same on every trial.
3. **(b)** — $P(X \geq 8) = P(X > 7) =$ `pbinom(7, 25, 0.2, lower.tail=FALSE)`.
4. A vertical line at your **observed count** x so you can compare data to the null distribution.
5. `dbinom` — one probability per bar; `pbinom` gives cumulative probabilities.
6. Example: If the true proportion were 10%, data this extreme would occur only about 3% of the time — evidence the proportion is **greater than** 10%.
